BUSINESS REQUIREMENTS DOCUMENT

TripLedger

Trip Budget & Expense Tracker

Offline-First Progressive Web App

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"Log a ₹800 fuel stop in Chitradurga with zero signal. Sync when you hit Hampi."

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1. Executive Summary

TripLedger is an offline-first Progressive Web App (PWA) purpose-built for the Indian road-trip traveller. It solves a single, deeply felt pain point: the inability to track shared trip expenses in areas with unreliable or absent mobile connectivity — a condition that defines the majority of scenic travel corridors across India (Hampi, Spiti, Coorg, Ladakh, and beyond).

> **Problem Statement:** Existing expense-tracking apps (Splitwise, TravelSpend, Trail Wallet) all require an active internet connection to log or sync data. In a country where 4G dead zones span hundreds of kilometres on high-value travel routes, these tools consistently fail at the exact moments they are needed most.

TripLedger resolves this by treating offline as the default state. The app is installed natively on a phone via PWA standards, stores all data locally via IndexedDB, and synchronises automatically — with full conflict resolution — the moment a connection is restored. No app store. No install friction. No data loss.

1.1 Product Vision

To become the go-to financial co-pilot for Indian road-trip groups — a product that is as reliable as the travellers themselves, even when the network is not.

1.2 Strategic Summary

Dimension	Summary
Target User	Indian road-trip travellers, 22–38 yrs, travelling in groups of 2–6
Core Differentiator	100% offline functionality with automatic, conflict-aware cloud sync
Technology Bet	PWA (Vite + React) + Dexie.js (IndexedDB) + Supabase + Workbox Service Worker
MVP Timeline	12 weeks from kickoff to public beta
Initial Markets	India (Phase 1); South East Asia (Phase 2 — Q3 FY27)
Monetisation	Freemium — free for solo trips; ₹199/trip for group sync (>2 devices)

2. Business Objectives

All objectives are time-boxed to 12 months post-public-beta launch unless stated otherwise. Each objective maps to a specific, measurable outcome and a responsible owner.

2.1 Primary Objectives

#	Objective	Measurable Target	Timeline	Owner
O-1	Acquire early adopter base through organic, word-of-mouth within the Indian travel community	5,000 active PWA installs within 90 days of beta launch	Q2 FY27	Growth / Marketing
O-2	Demonstrate product-market fit via retention	30-day retention rate ≥ 40% (industry avg. for utility apps: 25%)	Q3 FY27	Product
O-3	Validate freemium-to-paid conversion on group sync feature	≥ 8% of multi-device trips convert to paid (₹199/trip)	Q3 FY27	Revenue / Product
O-4	Establish offline reliability as the category-defining benchmark	Zero data loss incidents across 10,000 sync events (tracked via Sentry)	Q2–Q3 FY27	Engineering
O-5	Build brand presence in the Indian travel content ecosystem	3 partnerships with travel YouTubers/bloggers (>100K subscribers each)	Q3 FY27	Marketing

2.2 Success Metrics Dashboard

The following KPIs will be tracked weekly in a Posthog dashboard accessible to all stakeholders:

KPI	Baseline (Launch)	Target (M+6)	Source
PWA Installs	0	5,000	Vercel Analytics
D30 Retention	— (est. 20%)	≥ 40%	Posthog
Expenses Logged Offline (%)	—	≥ 60%	Supabase + Sentry
Sync Conflict Rate	—	< 2% of syncs	Sync Log Table
Paid Conversion (Group)	0%	≥ 8%	Stripe / Supabase
NPS Score	—	≥ 50	In-app Survey

3. Background & Context

3.1 Why Now?

Three converging macro trends create an exceptional window for this product in FY26–27:

- **Post-pandemic road trip boom: Domestic travel in India grew 32% YoY in 2024 (Ministry of Tourism data). Road trips specifically surged as travellers sought flexibility over fixed itineraries.**
- **PWA maturity: iOS 16.4+ (2023) added full PWA support including Background Sync and Push Notifications, removing the last major platform barrier. India's dominant Android base has had full PWA capability since 2019.**
- **Rural connectivity paradox: While urban 5G rollouts dominate headlines, 40%+ of India's scenic tourist corridors — NH-4 (Bangalore-Hampi), Manali-Leh, Spiti Valley — have documented dead zones exceeding 60km stretches (TRAI 2024 report).**

3.2 Competitive Landscape

No existing competitor treats offline-first as a core architectural constraint. All operate on an "online with fallback" model — the inverse of what the road-trip use case demands.

Feature	TripLedger	Splitwise	TravelSpend	Trail Wallet	Google Sheets
Offline logging	**Yes (full)**	No	Partial	Partial	No
Auto background sync	**Yes**	No	No	No	No
Conflict resolution UI	**Yes (HLC-based)**	No	No	No	Manual
PWA / No app store	**Yes**	No	No (iOS only)	No (iOS only)	Yes (browser)
Budget by category	**Yes**	No	Yes	Yes	Manual only
India pricing (INR)	**Yes (native)**	USD only	USD only	USD only	N/A

> **Strategic Gap:** TripLedger is the only product in this category designed for offline-first from the ground up — not as a feature, but as a non-negotiable architectural constraint. This is the moat.

3.3 User Research Insights

Qualitative interviews conducted with 24 road-trip travellers across Bangalore, Mumbai, and Delhi (Jan–Feb 2026) revealed the following:

- 18/24 had experienced data loss or app failure during a trip due to connectivity issues.
- 21/24 currently use WhatsApp group chats to manually track shared expenses — a high-friction workaround.
- 14/24 would pay up to ₹299/trip for a tool that "just worked in the hills."
- Top three requested features: (1) quick-add flow under 3 taps, (2) fuel cost splitting, (3) end-of-trip settlement summary.

4. Financials & ROI

4.1 Development Cost Estimate

All estimates assume a lean team operating on a 12-week MVP cycle with no infrastructure costs in the MVP phase (Vercel free tier + Supabase free tier handle early traffic).

Line Item	Unit Cost	Qty / Hrs	Total Estimate
Senior Full-Stack Engineer (contract)	₹3,500 / hr	240 hrs	**₹8,40,000**
UI/UX Designer (contract)	₹2,000 / hr	80 hrs	**₹1,60,000**
QA / Offline Testing (contract)	₹1,500 / hr	60 hrs	**₹90,000**
Product Management (internal)	—	80 hrs	**₹0 (internal)**
Infrastructure (Post-free-tier, Month 4+)	₹8,000 / mo	8 months	**₹64,000**
Marketing & Community Seeding	—	Lump sum	**₹1,20,000**
TOTAL YEAR-1 INVESTMENT			**₹12,74,000**

4.2 Revenue Projections (12-Month Model)

Conservative revenue model based on freemium group-sync monetisation at ₹199/trip:

Metric	Q1	Q2	Q3	Q4
Active PWA Installs	800	3,200	7,500	14,000
Multi-device Trips Created	120	640	1,800	3,500
Paid Conversion (8%)	10	51	144	280
Revenue @ ₹199/trip	₹1,990	₹10,149	₹28,656	₹55,720
CUMULATIVE REVENUE	**₹1,990**	**₹12,139**	**₹40,795**	**₹96,515**

ROI Note: Year-1 revenue (conservative: ~₹97K) does not offset the ₹12.74L investment. This is intentional — TripLedger is a category creation play. The business case rests on reaching 50,000 installs by Year 2 and introducing a ₹999/year premium tier (trip archive, PDF export, split settlement via UPI). Projected Year-2 revenue: ₹18–24L. Break-even at Month 22.

4.3 Cost Avoidance / Infrastructure Savings

- Vercel + Supabase free tiers handle the first 1,000 users at ₹0/month — no infrastructure spend for the first 90 days post-launch.
- PWA distribution eliminates the 30% Apple/Google Play revenue cut that would apply to a native app.
- Dexie.js + Service Worker caching reduces Supabase API calls by an estimated 70% (cached reads served from IndexedDB), directly reducing database compute costs at scale.

5. Stakeholders

5.1 Stakeholder Matrix

Stakeholders are categorised by influence (ability to affect the project) and interest (degree of involvement in outcomes).

Role	Name / Team	Responsibilities	Influence	Engagement Mode
Executive Sponsor	Founding Team / CEO	Final sign-off on scope, budget, and launch date	**High**	Monthly steering review
Product Manager	Senior PM (Author)	Requirements ownership, sprint prioritisation, stakeholder comms	**High**	Daily
Lead Engineer	Full-Stack Architect	Technical architecture, offline-sync engine, Supabase schema	**High**	Daily
UX Designer	Contract Designer	Mobile-first UI, offline state design, sync indicator patterns	**Medium**	Weekly design reviews
QA Engineer	Contract QA	Offline simulation testing, sync conflict scenarios, PWA install flows	**Medium**	Per sprint
Marketing Lead	Growth / Marketing	Community seeding, travel influencer partnerships, launch campaign	**Medium**	Bi-weekly
End Users (Beta)	Road-trip travel groups	Feature validation, bug reporting, NPS feedback	**Low–Medium**	In-app feedback + Discord
Legal / Compliance	External Counsel	Data privacy (DPDP Act 2023), Terms of Service, payment compliance	**Low**	Pre-launch review only

5.2 Sign-Off Requirements

The following sign-offs are required before each phase gate:

Gate	Sign-Off Required From	Criteria
BRD Approval	Executive Sponsor + PM	This document reviewed and signed
Architecture Sign-Off	Lead Engineer + PM	Tech stack finalised, data schema approved
Beta Launch	PM + Executive Sponsor	Core offline flow passing QA; Sentry integrated; 0 P1 bugs
Public Launch	All stakeholders	Legal review complete; payment integration live; NPS ≥ 40 from beta cohort

6. High-Level Scope

6.1 In Scope — MVP (Phase 1: Weeks 1–12)

Core Application

- PWA shell: installable on Android and iOS via browser 'Add to Home Screen' prompt
- Full offline operation: all read/write flows function without an internet connection
- IndexedDB storage via Dexie.js with reactive state management (Zustand)
- Workbox-powered Service Worker: app shell caching, background sync, precaching

Trip Management

- Create and manage trips: name, dates, base currency (INR default), total budget
- Define per-category budgets: Fuel, Food, Stay, Toll, Entertainment, Other
- Share trip via 6-digit trip code; co-travellers join and sync via Supabase

Expense Logging

- Quick-add flow: category → amount → optional note → done (≤ 3 taps)
- Optimistic UI: expense appears instantly without waiting for sync
- Client-side UUID generation to ensure ID integrity while offline
- HLC (Hybrid Logical Clock) timestamp on every record for conflict resolution

Budget Variance Dashboard

- Real-time variance view: spent vs. budgeted per category
- Progress bars with colour-coded health indicators (green/amber/red)
- Remaining budget and projected spend-rate based on trip duration

Sync Engine

- Background sync via Service Worker Sync API (retry on reconnect)
- HLC-based field-level conflict detection on the Supabase edge function
- Conflict resolution UI: user sees both versions and picks the winner
- Sync status indicator persistent in app header: Synced / Pending (N) / Offline / Conflict

6.2 Out of Scope — MVP

Out of Scope: The following items have been explicitly deferred to maintain a 12-week delivery timeline and avoid feature bloat diluting the core offline experience.

Feature	Rationale for Deferral	Target Phase
Receipt photo capture & storage	Adds file upload complexity; requires offline-to-cloud blob sync. Core value is financial tracking, not receipts.	Phase 2
Multi-currency with live exchange rates	INR handles 95%+ of target use cases in Phase 1. Live rate API adds online dependency.	Phase 2

GPS-based location tagging	Typed location notes are sufficient for MVP. GPS adds permission complexity and battery drain.	Phase 2
Charts & analytics (spend trends)	Post-trip analysis is a secondary use case. Core need is in-trip tracking.	Phase 2
PDF / CSV export	Valuable but not blocking the core job-to-be-done.	Phase 2
UPI-integrated settlement	Requires RBI-regulated payment gateway integration. Legal complexity.	Phase 3
Native iOS / Android app	PWA achieves equivalent functionality without app store dependency or ₹25K/yr developer fees.	Phase 3
International trip support (SE Asia)	Phase 2 market expansion once India PMF is validated.	Phase 2

7. Risks & Constraints

7.1 Risk Register

Each risk is rated by Probability (P) and Impact (I) on a 1–3 scale. Risk Score = P × I. Scores ≥ 4 are classified HIGH and require a mitigation plan before launch.

#	Risk	P (1–3)	I (1–3)	Score	Mitigation	Owner
R1	iOS PWA Background Sync limitation: iOS does not fully support the Background Sync API — sync only fires when app is open.	3	3	9 ★	Implement a foreground sync poller as fallback: check connection on app focus and trigger sync. Document limitation clearly to users in onboarding.	Engineering
R2	IndexedDB storage quota exceeded on low-end Android devices (often 50MB limit on budget phones).	2	3	6 ★	Implement LRU eviction for synced records older than 90 days. Show storage usage in Settings. Target: <5MB for 500 expenses.	Engineering
R3	Sync conflict rate higher than expected in real-world multi-device trips, leading to user frustration.	2	2	4 ★	Log all conflicts in Sentry from Day 1. If conflict rate >2% at Week 4 of beta, implement optimistic locking on the Supabase edge function.	Engineering + PM
R4	Supabase free tier limits (500MB DB, 2GB bandwidth) exceeded earlier than projected with viral growth.	2	2	4 ★	Set up Supabase usage alerts at 70% threshold. Pro plan ($25/mo) is the immediate upgrade path — trivial cost relative to user growth.	Engineering + Finance
R5	DPDP Act 2023 (India's data privacy law) compliance requirements for storing financial data.	2	2	4	Engage legal counsel pre-launch. Data stays in Supabase Mumbai region (ap-south-1). No PII beyond device ID and trip data. Privacy policy drafted.	PM + Legal
R6	Low organic discovery: PWA apps do not appear in app stores, limiting discoverability.	3	1	3	Seed in Indian travel communities (Thrillophilia, Zostel, IndieWanderer subreddit). Partner with 3 travel YouTubers for launch-day coverage.	Marketing

R7	12-week timeline slippage due to complexity of offline sync engine underestimation.	2	2	4	Sync engine is prioritised in Week 1–4. If behind by Week 6, cut the conflict resolution UI from MVP and ship with 'last write wins' as a safe fallback.	PM + Engineering

7.2 Technical Constraints

- **Service Worker scope: The SW can only control pages under its registered path. The entire app must be served from a single origin — no multi-domain setup.**
- **IndexedDB is synchronous-origin-bound: Data cannot be shared between browser tabs of different origins. Cross-device sync must go through Supabase.**
- **Background Sync API browser support: Fully supported on Chrome/Android. Not supported on Safari/iOS (see R1). Firefox support is partial.**
- **Supabase Row Level Security (RLS): All tables must have RLS policies enabled before launch. A misconfigured policy could expose one user's trip data to another.**

7.3 Regulatory & Compliance Constraints

- **India Digital Personal Data Protection Act (DPDP) 2023: User data must be stored in India. Supabase Mumbai (ap-south-1) region satisfies this. No international data transfer without explicit consent.**
- **Payment compliance: If UPI settlement is added in Phase 3, an aggregator licence via Razorpay or PhonePe is required. This is explicitly out of scope for MVP.**
- **No financial advice: The app tracks expenses only and must not display any content construable as financial or investment advice.**

7.4 Delivery Milestones

Week	Milestone	Deliverable	Sign-Off
1–2	Architecture & Schema Finalised	Dexie schema, Supabase schema, Zustand store skeleton	Lead Engineer + PM
3–5	Offline Expense Flow (Alpha)	Full CRUD against IndexedDB; PWA manifest + Workbox installed	PM
6–8	Cloud Sync Engine	Push/pull sync; HLC conflict detection; Sentry integrated	Lead Engineer + PM
9–10	Dashboard + Sync UI	Budget variance view; sync status indicator; conflict resolution screen	PM + Designer
11	QA & Beta Prep	Offline simulation tests; P1/P2 bug fix; beta invite list seeded	QA + PM

12	Beta Launch	Deployed to Vercel; 100-person closed beta; NPS baseline captured	All stakeholders

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